

MuJoCable Method and Implementation Report

MuJoCable

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Abstract

MuJoCable implements `mujoco.cable.unilateral`, a physics-informed, massless-cable surrogate for pulleys, winches, tensegrity, rolling-contact joints, hands, and routed compliant robots. It is not a deformable-rope solver: cable mass, sag, bending, torsion, waves, and self-contact remain outside its state. MuJoCo continues to own rigid-body dynamics, contact, constraints, and time integration.

The plugin preserves `route_mode="native"` and adds runtime surface routes over arbitrarily oriented cylinders and closed convex or guided nonconvex mesh obstacles. Initialization hints select wrap order and homotopy, but runtime length, velocity, contact nodes, tension, and body forces use the solved route. The constitutive model is unilateral, supports controlled free length and winch reserve, and optionally propagates segment tension using the Euler-Eytelwein relation. Persistent role-3 guides provide a reduced-order eyelet model with local friction. The latest release adds a dual-reserve, antagonistic log-spiral robot benchmark with module self-collision and a five-coefficient friction sweep.

1. Scope and MuJoCo Boundary

Native MuJoCo spatial tendons are preferred for routes explicitly defined by sites and supported wrap objects. MuJoCable targets mechanisms that also need:

1. initialization hints that do not remain fixed force nodes;
2. tangent/geodesic routes on moving obstacle surfaces;
3. unilateral slack with controlled free length;
4. direct surface-node force application without a second seed-tendon force;
5. route validity, residual, tension, and free-length diagnostics.

The division of responsibility is:

Layer	Owner	State and computation
Rigid mechanics	MuJoCo	bodies, joints, inertia, collision, constraints, integration
Route geometry	MuJoCable	tangent points, surface corridors, common-tangent bridges
Cable dynamics	MuJoCable	free length, slack, tension, friction, spool state
Rendering	MuJoCo scene + plugin callback	runtime cable polyline and state color

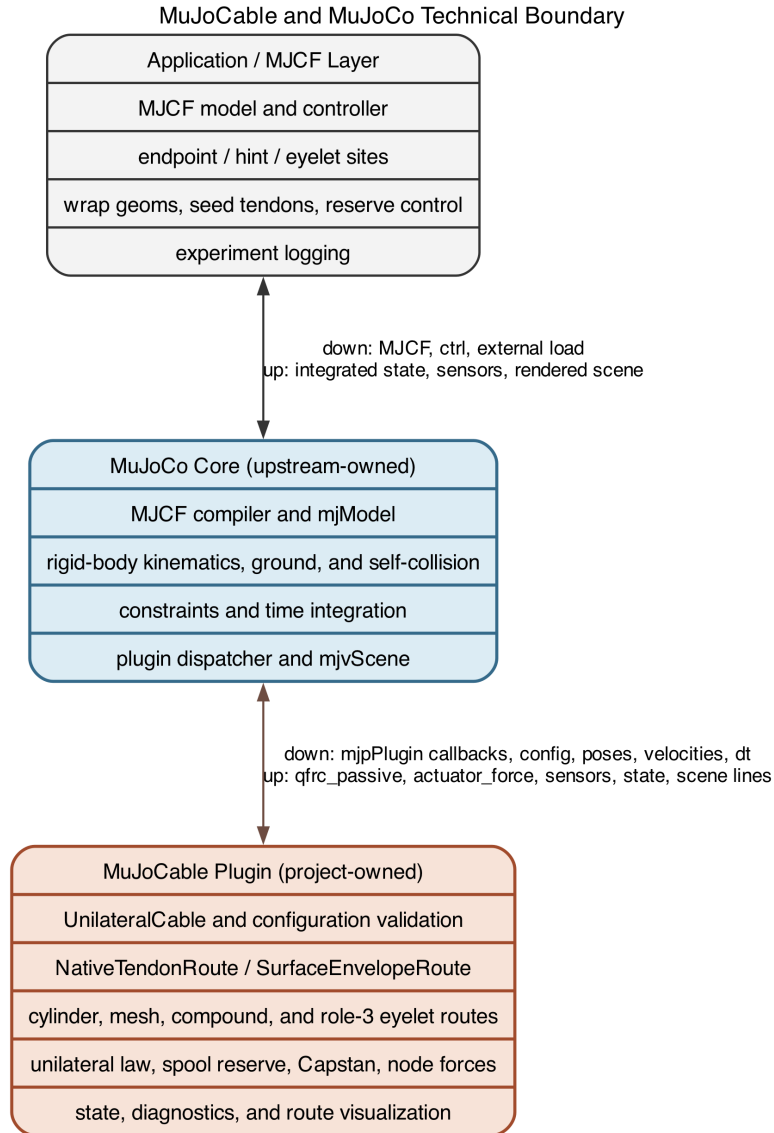


Figure 1: Component boundary

The plugin writes surface-route forces into `qfrc_passive`. A surface actuator keeps MuJoCo's control and force-reporting interface but must use `gear="0"`, preventing the force-disabled seed tendon from applying a duplicate transmission force.

2. MJCF Interface and Route Roles

The first site user field defines:

- user="1": physical endpoint;
- user="2": initialization hint, ignored as a runtime force node;
- user="3": persistent physical guide/eyelet;
- user="0": no plugin role.

```
<size nuser_site="1"/>
<tendon>
  <spatial name="route_seed" width="0.000000001">
    <site site="start"/><site site="proximal_hint"/>
    <site site="distal_hint"/><site site="end"/>
  </spatial>
</tendon>
<config key="route_mode" value="surface"/>
<config key="mesh_route_mode" value="guided_surface"/>
<config key="route_tendon" value="route_seed"/>
<config key="wrap_geoms" value="proximal_surface distal_surface"/>
```

Every role-2 hint must correspond to one wrap_geoms entry. The same obstacle may appear multiple times. A role-3 guide requires no wrap geom and remains a runtime force node.

The 12-value surface sensor returns

length, velocity, free_length, contraction, extension, tension,
taut, saturated, route_status, tangent_residual,
surface_residual, solver_iterations.

3. Route Solvers

3.1 Native backend

NativeTendonRoute reads MuJoCo's compiled spatial-tendon length, velocity, wrap points, and wrap ownership. The plugin adds unilateral constitutive and free-length control while retaining the native route.

3.2 Cylinder envelope

A cylinder is transformed to local coordinates. Entry and exit tangency, axial locations, and angular branch are optimized jointly. Unrolling the cylinder gives a straight geodesic, so a helical surface segment has length

$$L_s = \sqrt{R^2 \Delta\phi^2 + \Delta z^2}.$$

The total route combines endpoint tangent segments and the surface geodesic. The hint selects the angular branch; it does not appear in the final length.

3.3 Convex and guided nonconvex mesh routes

For a closed convex mesh, the hint selects a seed face and a triangle corridor. The strip is unfolded and edge-crossing coordinates are optimized for tangential continuity and minimum length.

For a closed nonconvex obstacle, the solver adds mesh adjacency and BVH visibility tests. A taut-obstacle route adds contact where a free segment penetrates and removes contacts requiring adhesive normal force, allowing a taut cable to bridge a concavity.

guided_surface preserves an initialized surface corridor for ligament-style routing. Multiple moving surfaces share a global length objective and a finite common-tangent bridge.

Meshes must be closed, manifold, consistently oriented, and non-self-intersecting. The current solver preserves the initialized topology and does not perform arbitrary side switching.

4. Cable Constitutive Law and Forces

For route length L , controlled contraction c , home length L_0 , pretension offset p , and deadband s ,

$$L_{free} = L_0 - p - c, \quad e = L - L_{free} - s.$$

The unilateral Kelvin-Voigt law is

$$T = \text{clip} \left(k[e]_+ + d\dot{L}, 0, T_{max} \right).$$

Slack cables therefore cannot push. Stiffness may be initialized from EA/L ; Poisson ratio is not an independent parameter in this one-dimensional model.

At each runtime node x_i , adjacent segment directions give

$$f_i = T_{i-1} \hat{t}_{i-1} - T_i \hat{t}_i.$$

The plugin applies each force at its world point using `mj_applyFT`. Virtual-work validation checks

$$q_{cable} \approx -T \frac{\partial L}{\partial q}.$$

Winch reaction

If contraction is coupled to a physical spool angle, enabling reaction torque adds

$$\tau_{spool} = -T \frac{dc}{d\theta}.$$

For two radii on a common shaft, this recovers the quasi-static wheel-and-axle balance $T_{in} R_{in} = T_{out} R_{out}$.

5. Friction and Eyelets

For wrap angle θ , the Euler-Eytelwein relation is

$$\frac{T_{high}}{T_{low}} = \exp(\mu\theta).$$

Surface nodes and role-3 guides propagate segment tension within the same force path. A guide-local forward relation is

$$T_{out} = T_{in} \exp(-\mu_g \theta_g).$$

Velocity-directed surface friction uses the regularization

$$\frac{T_{out}}{T_{in}} = \exp \left(\mu\theta \tanh(v_{slip}/v_0) \right),$$

so the direction reverses continuously with relative cable/surface sliding. This remains a kinetic-friction approximation without static-friction history, distributed bore contact, wear, or exact no-slip constraints.

6. Antagonistic Reserve and Log-Spiral Routing

When one side of an antagonistic mechanism contracts, the opposite route usually lengthens. A fixed opposite spool then creates unwanted antagonistic tension. With reserve R ,

$$L_{free} = L_{home} + R - c.$$

In MJCF, `pretension_offset=-R` implements the reserve. Demo 33 first takes up both sides and then performs differential payout:

$$(c_+, c_-) : (0, 0) \rightarrow (R, R) \rightarrow (R + u, R - u), \quad 0 \leq u \leq R.$$

The log-spiral geometry defines twelve progressively scaled modules. MuJoCo owns eleven hinges and module collision; MuJoCable owns the two threaded role-3 routes, free length, unilateral tension, and eyelet friction. Eleven explicit adjacent contact pairs supplement ordinary mesh collision.

For $R=25$ mm, active/release commands of $50/0$ mm produced zero final release-side tension and approximately 176.8 deg net bend. The very small active tension (0.061 N) reveals under-calibrated hinge stiffness, so this is a route/control validation rather than a hardware-accurate prediction.

Eyelet friction μ	Net bend (deg)	Active tension (N)	Release tension (N)
0	177.54	0.0588	0
0.015	176.83	0.0612	0
0.05	175.24	0.0677	0
0.10	173.10	0.0802	0
0.20	168.88	0.1257	0

7. Numerical Stabilization

Control filtering, maximum contraction rate, taut/slack hysteresis, route hysteresis, and visual smoothing reduce switching jitter without changing the static law. The optional matrix-free implicit-compliance mode linearizes a fixed route and taut active set:

$$A = M + \sum_i (h d_i + h^2 k_i) G_i^T G_i.$$

It improves high-stiffness cable oscillation but cannot remove changes caused by route-corridor or contact-active-set switching; those require geometric and contact hysteresis.

8. Validation Summary

- single- and dual-cylinder length errors: below $1e-6$ m;
- cylinder tangent residuals: below $5e-8$;
- virtual-work absolute error: $1.43e-5$ N;
- frictionless endpoint tension ratio: 1.0 ;
- $\mu=0.35$ measured/discrete Capstan ratios: $1.1811301/1.1811306$;
- slack endpoint force: 0 N;

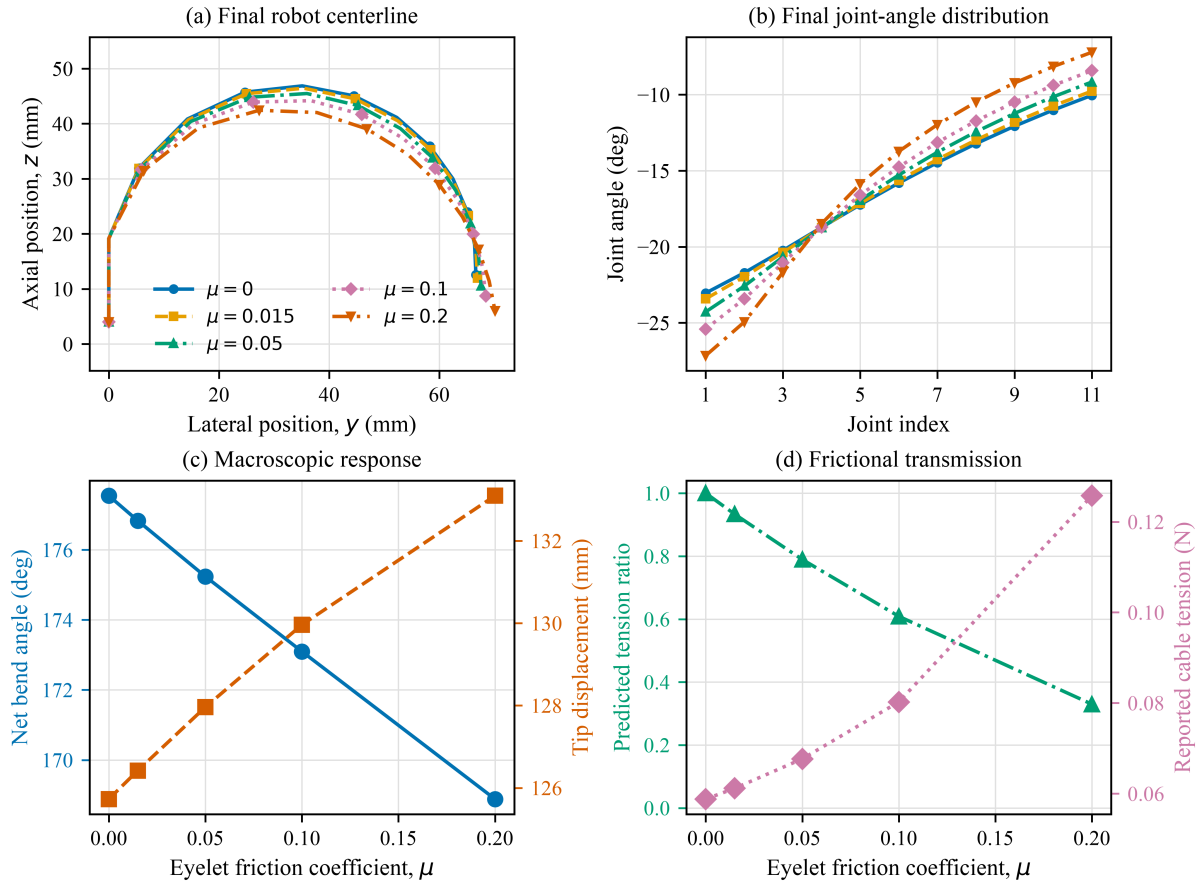


Figure 2: Dual-reserve friction sweep

- wheel-and-axle theoretical/measured tension ratios: $3.000/2.995$;
- free inertial pulley peak angular speed: 8.65 rad/s ;
- free-pulley regularized Capstan p95 relative error: $2.69\text{e-}4$;
- Demo 33 initial/maximum module contacts: $4/4$;
- Demo 33 maximum numerical penetration: approximately $4.2\text{e-}9 \text{ m}$;
- all five Demo 33 friction cases ended with zero release-side tension.

These are simulation verification results, not experimental identification. Hardware validation should measure route shape, both-side tension, spool angle, payload motion, and energy loss over held-out speeds and friction conditions.

9. Limitations

1. The cable is massless and has no sag, bending, torsion, waves, cable-cable contact, or material transport.
2. Mesh routes retain initialized homotopy and do not globally change sides.
3. Finite-cylinder end-face transitions are not implemented.
4. Guided eyelets are point models without bore geometry, jamming, or wear.
5. Capstan friction is regularized sliding friction without stick-slip history.
6. Surface forces use `qfrc_passive`; actuator transmission is retained only for control and force reporting.
7. Robot mesh self-collision is a MuJoCo MJCF responsibility, not a cable plugin capability.
8. The models are physics-informed mechanism surrogates, not validated digital twins unless their geometry and material parameters are experimentally identified.

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